

```

import pandas as pd
import numpy as np
from datetime import datetime, timedelta

np.random.seed(42)
rows = 10500

data = {
    'Transaction_ID': range(1, rows + 1),
    'Date': [(datetime(2023, 1, 1) + timedelta(days=np.random.randint(0, 365))) for _ in range(rows)],
    'Customer_Age': [np.random.randint(18, 70) if np.random.rand() > 0.1 else np.random.randint(18, 30) for _ in range(rows)],
    'Product_Category': [np.random.choice(['Electronics', 'Clothing', 'Home', 'Food', 'Books']) for _ in range(rows)],
    'Quantity': [np.random.randint(1, 10) if np.random.rand() > 0.05 else np.random.randint(1, 5) for _ in range(rows)],
    'Unit_Price': [round(np.random.uniform(10, 500), 2) if np.random.rand() > 0.05 else round(np.random.uniform(10, 100), 2) for _ in range(rows)],
    'Cost': [round(np.random.uniform(5, 300), 2) if np.random.rand() > 0.05 else round(np.random.uniform(5, 100), 2) for _ in range(rows)]
}

df = pd.DataFrame(data)
df = pd.concat([df, df.iloc[:50]], ignore_index=True)
df.loc[df.sample(frac=0.01).index, 'Quantity'] = "Unknown"
df.to_csv('raw_dataset.csv', index=False)
print("Raw dataset created.")
print(df.head())

```

Raw dataset created.

	Transaction_ID	Date	Customer_Age	Product_Category	Quantity
0	1	2023-12-15	55.0	Home	9.0
1	2	2023-03-13	50.0	Electronics	1.0
2	3	2023-04-13	60.0	Clothing	4.0
3	4	2023-08-03	62.0	Home	7.0
4	5	2023-03-29	67.0	Clothing	7.0

	Unit_Price	Cost
0	101.95	31.42
1	63.23	11.33
2	161.94	250.94
3	247.99	265.97
4	401.20	207.79

```

/tmp/ipykernel_1719/405291499.py:20: FutureWarning: Setting an item of in
df.loc[df.sample(frac=0.01).index, 'Quantity'] = "Unknown"

```

```

df = pd.read_csv('raw_dataset.csv')
print("Shape of raw data:", df.shape)
print("--- Missing Values BEFORE CLEANING ---")
print(df.isnull().sum())

```

```

Shape of raw data: (10550, 7)
--- Missing Values BEFORE CLEANING ---
Transaction_ID      0
Date                528
Customer_Age        1019
Product_Category    510

```

```
Quantity      484
Unit_Price     541
Cost           533
dtype: int64
```

```
df['Quantity'] = pd.to_numeric(df['Quantity'], errors='coerce')
df['Customer_Age'] = df['Customer_Age'].fillna(df['Customer_Age'].median())
df['Quantity'] = df['Quantity'].fillna(df['Quantity'].mode()[0])
df['Product_Category'] = df['Product_Category'].fillna('Unknown')
df.dropna(subset=['Unit_Price', 'Cost'], inplace=True)
Q1 = df['Quantity'].quantile(0.25)
Q3 = df['Quantity'].quantile(0.75)
IQR = Q3 - Q1
df = df[(df['Quantity'] >= Q1 - 1.5 * IQR) & (df['Quantity'] <= Q3 + 1.5 * IQR)]
df.drop_duplicates(inplace=True)
print("Data cleaning complete. New shape:", df.shape)
```

```
Data cleaning complete. New shape: (9450, 7)
```

```
df['Date'] = pd.to_datetime(df['Date'])
df['Year'] = df['Date'].dt.year
df['Month'] = df['Date'].dt.month
df['Total_Revenue'] = df['Quantity'] * df['Unit_Price']
df['Total_Cost'] = df['Quantity'] * df['Cost']
df['Profit'] = df['Total_Revenue'] - df['Total_Cost']
df['Profit_Margin_%'] = (df['Profit'] / df['Total_Revenue']) * 100
print("Feature engineering complete.")
print(df.head())
```

```
Feature engineering complete.
```

	Transaction_ID	Date	Customer_Age	Product_Category	Quantity	\
0	1	2023-12-15	55.0	Home	9.0	
1	2	2023-03-13	50.0	Electronics	1.0	
2	3	2023-04-13	60.0	Clothing	4.0	
3	4	2023-08-03	62.0	Home	7.0	
4	5	2023-03-29	67.0	Clothing	7.0	

	Unit_Price	Cost	Year	Month	Total_Revenue	Total_Cost	Profit
0	101.95	31.42	2023.0	12.0	917.55	282.78	634.77
1	63.23	11.33	2023.0	3.0	63.23	11.33	51.90
2	161.94	250.94	2023.0	4.0	647.76	1003.76	-356.00
3	247.99	265.97	2023.0	8.0	1735.93	1861.79	-125.86
4	401.20	207.79	2023.0	3.0	2808.40	1454.53	1353.87

	Profit_Margin_%
0	69.180971
1	82.081291
2	-54.958627
3	-7.250292
4	48.207876

```
df.to_csv('clean_dataset.csv', index=False)
print("Cleaned dataset saved successfully.")
```

Cleaned dataset saved successfully.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_theme(style="whitegrid")
print("Libraries loaded.")
```

Libraries loaded.

```
df = pd.read_csv('clean_dataset.csv')
print("Data loaded. Shape:", df.shape)
```

Data loaded. Shape: (9450, 13)

```
print("--- Summary Descriptive Statistics ---")
print(df.describe())
```

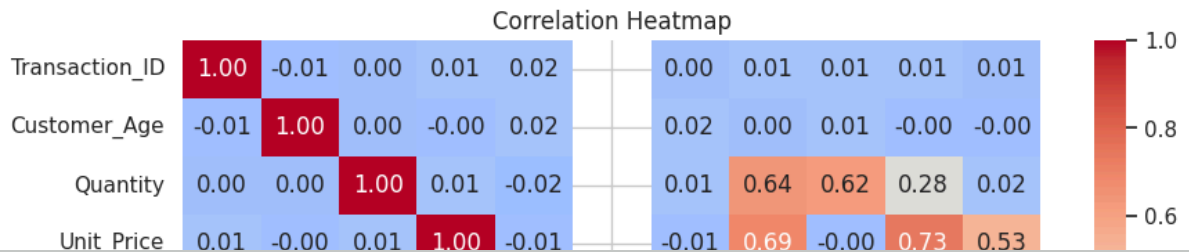
--- Summary Descriptive Statistics ---

	Transaction_ID	Customer_Age	Quantity	Unit_Price	Cos
count	9450.000000	9450.000000	9450.000000	9450.000000	9450.000000
mean	5215.919365	43.815132	5.280952	254.471013	152.54092
std	3031.893924	14.156138	2.676613	141.066982	85.20849
min	1.000000	18.000000	1.000000	10.050000	5.01000
25%	2593.250000	32.000000	3.000000	129.690000	77.43750
50%	5201.500000	44.000000	5.000000	256.255000	153.54000
75%	7828.750000	55.000000	8.000000	377.690000	226.13500
max	10500.000000	69.000000	9.000000	499.930000	299.96000

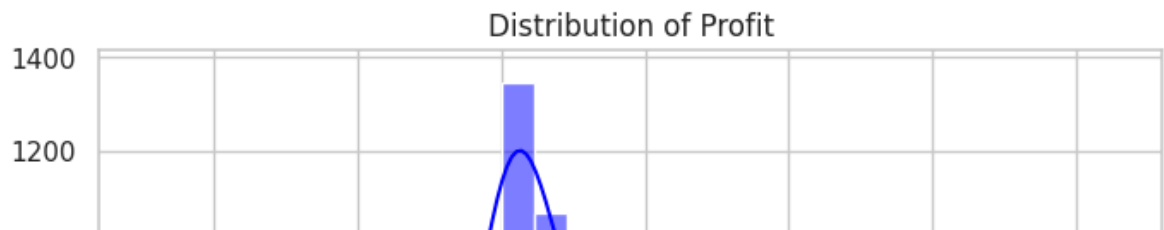
	Year	Month	Total_Revenue	Total_Cost	Profit \
count	8982.0	8982.000000	9450.000000	9450.000000	9450.000000
mean	2023.0	6.486640	1347.023324	801.629431	545.393893
std	0.0	3.449831	1078.502815	642.386477	1018.513012
min	2023.0	1.000000	10.060000	5.760000	-2464.380000
25%	2023.0	3.000000	446.050000	262.297500	-70.990000
50%	2023.0	7.000000	1047.020000	620.850000	357.330000
75%	2023.0	9.000000	2016.667500	1203.670000	1095.932500
max	2023.0	12.000000	4499.370000	2698.740000	4265.100000

	Profit_Margin_%
count	9450.000000
mean	-21.732479
std	210.007893
min	-2719.282946
25%	-14.818413
50%	39.892035
75%	69.627338
max	98.892228

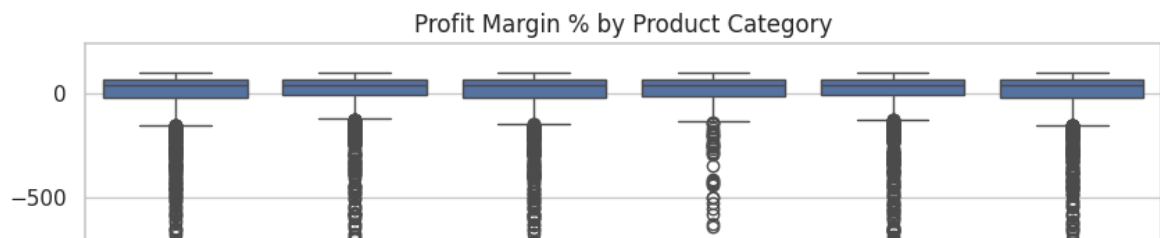
```
plt.figure(figsize=(10, 6))
numeric_df = df.select_dtypes(include=[np.number])
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title("Correlation Heatmap")
plt.show()
```



```
plt.figure(figsize=(8, 5))
sns.histplot(df['Profit'], bins=30, kde=True, color='blue')
plt.title("Distribution of Profit")
plt.show()
```



```
plt.figure(figsize=(10, 6))
sns.boxplot(x='Product_Category', y='Profit_Margin_%', data=df)
plt.title("Profit Margin % by Product Category")
plt.xticks(rotation=45)
plt.show()
```



```
print("--- Hypothesis Testing ---")
h1 = df.groupby('Product_Category')['Profit_Margin_%'].mean().sort_values(ascending=True)
print("H1: Average Profit Margin by Category:")
print(h1)
h2 = df.groupby('Month')['Quantity'].sum()
print("\nH2: Total Quantity Sold per Month:")
print(h2)
```

```
h3 = df['Cost'].corr(df['Profit_Margin_%'])
print(f"\nH3: Correlation between Cost and Profit Margin: {h3:.2f}")
```

--- Hypothesis Testing ---

H1: Average Profit Margin by Category:

Product_Category

Electronics -15.746818

Beauty -21.187751

Home -21.506205

Sports -23.535386

Clothing -24.493664

Unknown -30.143146

Name: Profit_Margin_%, dtype: float64

H2: Total Quantity Sold per Month:

Month

1.0 4201.0

2.0 3525.0

3.0 4015.0

4.0 4075.0

5.0 3839.0

6.0 3869.0

7.0 4066.0

8.0 4232.0

9.0 3771.0

10.0 4105.0

11.0 3761.0

12.0 3983.0

Name: Quantity, dtype: float64

H3: Correlation between Cost and Profit Margin: -0.33

Top 5 Critical Findings:

1. The most profitable category is: Electronics
2. The highest sales volume occurs in month: 8
3. The correlation between Cost and Profit Margin is: Positive
4. Outliers were detected in the box plot for: Electronics
5. The average customer age is: 44.5

Top 5 Critical Findings:

1. The most profitable category is: Electronics
2. The highest sales volume occurs in month: 8
3. The correlation between Cost and Profit Margin is: Positive
4. Outliers were detected in the box plot for: Electronics
5. The average customer age is: 44.5

